

IRS-Assisted UAV Secure Communication via Agentic AI

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Abstract—Low-altitude uncrewed aerial vehicle (UAV) communication offers notable advantages over terrestrial base stations in terms of flexibility and deployment efficiency. However, the high likelihood of line-of-sight (LoS) propagation renders the communication links between UAVs and ground users (GUs) particularly susceptible to eavesdropping. To address this issue, we consider an intelligent reflecting surface (IRS)-assisted low-altitude UAV secure communication system, in which communication security is strengthened through adaptive control of the wireless propagation environment, even when eavesdroppers are present. We aim to maximize the secrecy rate of GUs while minimizing the UAV energy consumption by jointly optimizing the continuous UAV trajectory, power allocation, and discrete IRS phase shifts. Considering the dynamic, non-convex, and NP-hard nature of the optimization problem, we propose an agentic artificial intelligence (AI) approach, namely alternating optimization (AO) and generative diffusion model-based deep deterministic policy gradient (AO-GDMDDPG) approach. The proposed agentic AI approach is composed of two cooperative agents that operate over a hybrid and high-dimensional decision space, in which the UAV agent adopts a generative AI (GenAI)-enhanced deep reinforcement learning (DRL) method to optimize continuous decision variables, whereas the IRS agent relies on the AO method to determine discrete IRS phase shifts. Simulation results demonstrate the superiority of the AO-GDMDDPG approach over benchmark algorithms with respect to secrecy rate improvement and UAV energy consumption reduction.

Index Terms—Agentic AI, IRS, UAV communications, physical layer security, DRL.

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I. INTRODUCTION

UNCREWED aerial vehicles (UAVs) have gained popularity for their high mobility and cost-effectiveness, and they have served various fields of low-altitude wireless networks (LAWN) [1], [2]. In particular, compared to fixed terrestrial base stations, UAVs offer superior three-dimensional mobility, making UAV-assisted communication systems more flexible and efficient in dynamic and complex environments [3]. However, UAV communication systems are susceptible to eavesdropping due to the inherently shared characteristics of wireless transmissions [4]. Specifically, eavesdroppers can intercept links between UAVs and legitimate users to illegally access sensitive information, which degrades communication confidentiality and reduces overall system reliability.

Another challenge facing low-altitude UAV systems is maintaining considerable transmission performance in complex terrain, especially in densely populated urban areas or mountainous terrain [5]. Specifically, line-of-sight (LoS) links between UAVs and ground users (GUs) are often blocked by tall buildings and terrain variations, causing significant path loss, multipath fading, and non-line-of-sight (NLoS) effects during signal propagation [6]. These factors sharply reduce the received signal strength and increase the risk of link outages. Therefore, it is crucial to implement novel methods that modify the characteristics of wireless signal propagation to enhance security and reliability.

Intelligent reflecting surface (IRS) can be seen as a promising solution to shape wireless propagation and establish a programmable radio environment [7]. Specifically, an IRS is composed of numerous passive reflecting elements, each adjustable in both phase and amplitude, enabling the passive reflection of incoming signals [8]. Moreover, due to its energy-green, low cost, and flexible deployment, IRS demonstrates more significant advantages over traditional physical-layer security methods [9]. As such, with the assistance of the IRS, we can avoid the negative influence of eavesdroppers and obstacles, thereby achieving secure and efficient communication between the UAV and GUs [10].

However, designing an effective optimization approach for the IRS-assisted UAV secure communication system presents several challenges. *First*, the high mobility of the UAV and the rapid variation of wireless channels introduce strong environmental dynamics, which require the optimization approach

to promptly adapt to time-varying conditions. *Second*, the IRS phase shifts should be carefully optimized to steer the reflected signals toward legitimate GUs for secure transmission enhancement. However, the large number of passive reflecting elements in the IRS results in a high-dimensional optimization space. *Finally*, the optimization approach should jointly consider instantaneous performance and long-term system performance to maintain reliable operation over time. As such, addressing the above challenges requires developing a novel method that can deal with the high dynamics, high dimensionality, and long-term dependency within the considered system.

Accordingly, we propose an agentic artificial intelligence (AI) approach to optimize the security performance and energy efficiency of the IRS-assisted low-altitude UAV secure communication system. The proposed agentic AI approach leverages the advantages of deep reinforcement learning (DRL) for dynamic sequential decisions and the ability of generative AI (GenAI) to model complex data distributions, allowing it to attain high decision accuracy in dynamic and complex systems [11], [12]. The main contributions are summarized as follows:

- *IRS-assisted Low-altitude UAV Secure Communication System:* We consider an IRS-assisted low-altitude UAV secure communication system in which a low-altitude UAV uses the non-orthogonal multiple access (NOMA) protocol for simultaneous communication with multiple legitimate users affected by obstacles while multiple eavesdroppers are present. Moreover, we deploy an IRS in the system to reconfigure the wireless propagation environment, thereby mitigating the severe path loss caused by obstacle blockage and preventing unauthorized eavesdropping. The considered system has practical applications, such as in emergency rescue scenarios, where ground infrastructure is typically damaged and sensitive information needs to be protected from eavesdropping.
- *Formulations of Multi-Objective Dynamic Optimization Problem:* To reduce information leakage while conserving the limited onboard energy of the UAV platform for improving the system sustainability, we formulate an IRS-UAV secure communication multi-objective optimization problem (IUSCMOP) that simultaneously optimizes UAV trajectory design, power allocation for legitimate GUs, and IRS phase shift control for maximizing the average secrecy rate and minimizing the average UAV energy consumption. Since the IUSCMOP is dynamic, non-convex, and NP-hard, conventional optimization methods face significant difficulties in addressing it effectively.
- *Agentic AI Approach:* We propose an agentic AI approach called alternating optimization and generative diffusion model-based deep deterministic policy gradient (AO-GDMDDPG), which effectively overcomes the limitations of conventional optimization methods. Specifically, we solve the complex IUSCMOP by assigning different agents to optimize the hybrid variables, thereby improving the overall solution quality. First, the IRS agent applies an effective alternating optimization (AO) method to determine the discrete IRS phase shifts. Subsequently,

the UAV agent employs a generative diffusion model-based deep deterministic policy gradient (GDMDDPG) to optimize continuous decision variables, where the generative diffusion model (GDM) is introduced to improve the environment state analysis capability and decision generation quality of the deep deterministic policy gradient (DDPG).

- *Simulations and Analyses:* Simulation results indicate that our agentic AI approach attains substantial gains in secrecy rate and energy efficiency while consistently outperforming a variety of comparison algorithms. In particular, the proposed agentic AI approach maintains outstanding and stable performance under different system settings, including variations in operation time lengths and IRS configurations, which demonstrates its robustness and adaptability to varying and complex wireless communication environments.

The rest of the paper is organized as follows. Section II gives the related work. Section III presents the system model and formulates the optimization problem. Section IV proposes the AO-GDMDDPG approach. Section V shows the simulation results. Finally, Section VI concludes the overall paper.

II. RELATED WORK

In this section, we review the existing studies in UAV secure communication systems. Table I compares our work with existing studies.

A. UAV Secure Communication Systems

As the low-altitude economy expands, the physical layer security of UAV communications has received increasing attention due to operational safety considerations. In this case, some existing studies focus on traditional security methods such as collaborative beamforming, friendly interference, and artificial noise (AN) to ensure the UAV communication security when eavesdroppers are present. For example, Sun et al. [13] used several UAVs to create a virtual antenna array and aligned the mainlobe toward the target base station to reduce information leakage and achieve secure data transmission. Moreover, Huang et al. [14] employed collaborative beamforming and introduced friendly jamming UAVs that transmit interference signals toward remote eavesdroppers to support secure maritime communications, thereby improving the robustness of secrecy transmission under challenging maritime conditions. In addition, Zhang et al. [15] used AN to interfere with illegal terrestrial detectors, which enables the covert communication process between the UAV and GUs. However, the aforementioned physical layer security methods exhibit several notable limitations. Specifically, collaborative beamforming is not suitable for single-UAV communication scenarios, while friendly jamming and AN experience performance degradation in environments with dense obstacles due to severe blockage.

To address the issues above, the IRS has become a promising solution for supporting UAV secure communication, as it is capable of reconfiguring the propagation environment. For example, Zhang et al. [16] leveraged an IRS to assist the

TABLE I
COMPARISON OF RELATED WORKS

Related works	Physical security method	Multiple access method	Optimization variables	Optimization objectives	Optimization approach
[13]	Collaborative beamforming	TDMA	UAV deployments, excitation current weights, BS selection	Secrecy rate, sidelobe level, UAV energy consumption	Swarm intelligence
[14]	Collaborative beamforming, friendly jamming	TDMA	UAV deployments, excitation current weights	Secrecy rate, eavesdropping rate, UAV energy consumption	Swarm intelligence
[15]	AN	TDMA	UAV trajectory, transmit power	Covert transmission rate	Successive convex approximation (SCA)
[16]	IRS beamforming	TDMA	Transmit power, user and target scheduling, IRS phase shifts, UAV trajectory	Average achievable rate	AO, SCA, manifold optimization
[17]	IRS beamforming	TDMA	Transmit power, IRS phase shifts, UAV trajectory	Maximum worst-case downlink secrecy rate	SCA, S-procedure, semi-definite relaxation
[18]	Relay	Not mentioned	Transmit power, UAV trajectory	Sum secrecy capacity	Block coordinate descent (BCD), SCA
[19]	IRS beamforming	TDMA	IRS phase shifts, UAV trajectory	Average secrecy rate	DRL
[20]	Relay	NOMA	Base station, UAV, and satellite beamforming, UAV trajectory	Worst-case secrecy rate	S-procedure, general sign-definiteness, SCA
[21]	AN	NOMA	CPU frequency, transmit power, beamforming vector, AN vector, UAV trajectory	Energy Consumption	Semi-definite relaxation, S-Procedure, first-order approximation, SCA
[22]	IRS beamforming	TDMA	UAV trajectory, IRS phase shifts	UAV energy consumption	Model predictive control
[23]	AN	NOMA	UAV trajectory, beamforming vector, and AN vector	UAV path, power consumption, secure communication rate	Model predictive control
[24]	Collaborative beamforming	TDMA	UAV deployments, excitation current weights, BS selection	Total secrecy rates, total maximum sidelobe level, total UAV energy consumption	Swarm intelligence
[25]	IRS beamforming	TDMA	UAV trajectory, UAV beamforming, IRS phase shifts	Average secrecy rate, secrecy outage duration	DRL
[26]	IRS beamforming	TDMA	UAV beamforming, UAV trajectory, IRS phase shifts	Secrecy rate	DRL
Our work	IRS beamforming	NOMA	IRS phase shifts, UAV trajectory, transmit power	Secrecy rate, UAV energy consumption	Agentic AI

UAV in secure communication with GUs and secure sensing of ground targets while an eavesdropper with unknown channel conditions is present. Moreover, Liu et al. [17] investigated a UAV-mounted IRS to improve downlink transmission security for the base station, which offers more flexible connectivity through mobile IRS deployment. However, the aforementioned studies only considered a single eavesdropper and adopted time division multiple access (TDMA) to allocate communication resources, which limits their applicability in scenarios involving multiple eavesdroppers and requiring communication efficiency.

B. Performance Metrics in UAV Secure Communication Systems

In UAV secure communication systems, improving communication security is the primary objective of most existing studies. For example, Tian et al. [18] aimed to increase the total secrecy capacity through the joint adjustment of transmit power and UAV trajectory in a relay UAV-based secure communication system. Moreover, Tang et al. [19] deployed both fixed IRS and UAV-mounted IRS to guarantee system security and aimed to improve the average secrecy rate during UAV

flight through the joint configuration of all IRS phase shifts and the UAV trajectory. In addition, Yao et al. [20] used a relay UAV to provide secure links connecting the terrestrial base station with remote maritime users, simultaneously adjusting the beamforming configuration and UAV trajectory to enhance the minimum secrecy rate.

Furthermore, given that UAVs are airborne platforms with limited onboard energy, minimizing UAV power usage is essential to extend system operation, especially in scenarios where recharging is inconvenient. For example, Mao et al. [21] introduced a UAV-assisted secure edge computing framework and designed a convex optimization method to reduce energy usage while meeting the secure transmission demand. Moreover, Huang et al. [22] studied a secure communication scheme supported by an IRS for UAV and mobile GUs, and they aimed to minimize UAV energy consumption when the achievable secrecy rate is satisfied. However, the aforementioned studies addressed either secrecy performance or UAV energy usage independently. Notably, optimizing secrecy rate alongside reducing UAV energy usage is critical for enhancing the long-term sustainability of UAV communication systems.

C. Optimization Approaches in UAV Secure Communication Systems

Various optimization algorithms have been applied to improve the performance of UAV secure communication systems, where several conventional optimization methods have received extensive attention due to their simple implementation. For example, Li et al. [23] proposed a model predictive control-based algorithm to decouple UAV trajectory, beamforming matrix, and AN vector for maximizing secrecy rate. Moreover, Li et al. [24] designed a collaborative beamforming scheme to reduce information leakage and proposed a multi-objective dragonfly algorithm to improve secrecy rate while reducing sidelobe level and energy consumption. However, the aforementioned conventional optimization methods pose challenges in handling highly dynamic environment characteristics. This is because they rely heavily on prior knowledge and iterative procedures, which limit adaptability in time-varying UAV secure communication systems.

In this case, DRL algorithms show significant advantages since they are well-suited to address dynamic optimization problems through the trial-and-error learning mechanism. For instance, Dong et al. [25] developed a DRL method based on proximal policy optimization (PPO) to simultaneously configure IRS phase settings and design the UAV trajectory for only maximizing communication performance in the IRS-assisted UAV communication system with a single user and eavesdropper. Moreover, Cao et al. [26] employed a hierarchical twin delayed deep deterministic policy gradient algorithm to perform hierarchical optimization of beamforming matrices and UAV trajectory, thereby improving the stability of the optimization process. However, DRL algorithms typically experience reduced convergence accuracy and slower convergence when addressing optimization problems with high-dimensional action spaces. In addition, their ability to analyze environmental states and generate high-quality actions is also limited in complex environments. Consequently, solving the optimization challenge in complex IRS-assisted UAV secure communication systems remains difficult when relying solely on conventional DRL algorithms.

D. Summary

Overall, the differences between our work and prior studies can be outlined as follows. *First*, many previous studies neglected the impact of obstacles on the UAV secure communication process. *Second*, many existing studies mainly focused on improving communication security while overlooking the critical influence of UAV energy consumption on system sustainability. *Third*, prior studies employed conventional optimization algorithms and DRL algorithms to address optimization problems in UAV secure communication systems, while these algorithms cannot simultaneously handle the dynamic, complex, and high-dimensional characteristics of the environment states and decision space. In contrast, our work considers an IRS-assisted low-altitude UAV secure communication system that employs NOMA protocol to improve communication efficiency, and we propose an agentic AI

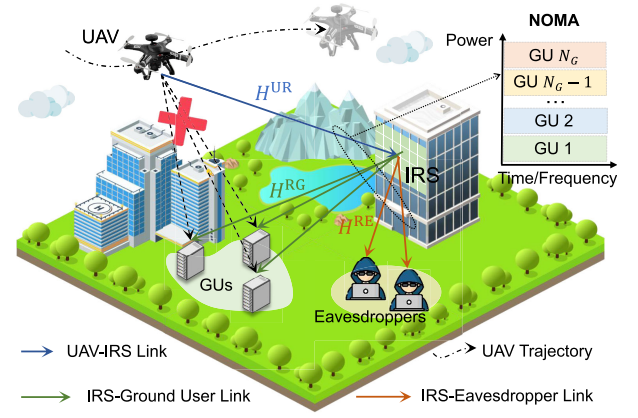


Fig. 1. IRS-assisted low-altitude UAV secure communication with NOMA in the presence of eavesdroppers.

approach aimed at simultaneously improving the secrecy rate and reducing UAV energy usage.

III. SYSTEM MODEL AND PROBLEM FORMULATION

This section presents a comprehensive analysis of the system model and the formulation of the optimization problem.

A. System Overview

As shown in Fig. 1, we consider an IRS-assisted low-altitude UAV communication system, where a UAV simultaneously serves a set of GUs denoted as $\mathcal{N} \triangleq \{1, \dots, n, \dots, N_G\}$ by using NOMA with the assistance of an IRS. The presence of environmental obstacles prevents the establishment of direct communication paths between the UAV and the GUs. As such, the UAV enables communications by sending signals to the IRS, which subsequently redirects them to the GUs via three-dimensional (3D) passive beamforming. Meanwhile, there exists a set of eavesdroppers represented by $\mathcal{K} \triangleq \{1, \dots, k, \dots, N_E\}$ whose objective is to intercept sensitive information conveyed through these communication links. We consider that the IRS consists of an array of $M_r \times M_c$ reflecting elements with d_r spacing in each row and d_c spacing in each column. Moreover, the UAV is considered to perform tasks at a fixed altitude H^U and is limited to a rectangular area.

Remark 1: Traditional and common orthogonal multiple access (OMA) protocols have limitations in our considered model. For example, TDMA has low transmission efficiency, and orthogonal frequency division multiple access (OFDMA) demonstrates low spectrum utilization efficiency. As such, we adopt the NOMA protocol to design the transmission process of the considered system, since it provides significant advantages in communication efficiency and spectrum utilization for multi-user secure communication tasks [27]. Nevertheless, the considered system is inherently extensible to other communication protocols, while the effectiveness of the optimization approach proposed later remains applicable.

A discrete time model is adopted where the overall duration T_f is uniformly divided into T equal intervals, indexed by $\mathcal{T} \triangleq \{1, \dots, t, \dots, T\}$. Meanwhile, the spatial distribution of

TABLE II
MAJOR NOTATIONS

Notation	Definition
$a^x[t], a^y[t]$	Horizontal displacement of the UAV along the x and y axes during time slot t
$d^{\text{UR}}[t]$	Distance from the UAV and IRS in time slot t
d_n^{RG}	Distance from the UAV and n -th GU
d_r, d_c	The interval between reflecting elements along each row and column, respectively
$d(i)$	Index of the GU ranked in descending order of channel quality
$e[t]$	UAV energy consumption in time slot t
$g_L^{\text{UR}}[t]$	LoS component of the channel between the UAV and IRS in time slot t
$g_{n,L}^{\text{RG}}[t], g_{n,N}^{\text{RG}}[t]$	LoS and NLoS components of the channel between the IRS and n -th GU in time slot t , respectively
$H^{\text{UR}}[t]$	Channel response vector between the UAV and IRS in time slot t
$H_n^{\text{RG}}[t], H_k^{\text{RE}}[t]$	Channel response vectors between the IRS and n -th GU and k -th eavesdropper in time slot t , respectively
$H_n^{\text{UG}}[t]$	Channel between the UAV and n -th GU in time slot t
$H_k^{\text{UE}}[t]$	Channel between the UAV and k -th eavesdropper in time slot t
$k, N_E, \mathcal{K}$	Index, total number, and set of eavesdroppers
M_r, M_c	Total reflecting elements along each row and column, respectively
$n, N, \mathcal{N}$	Index, total number, and set of GUs
P^{U}	Transmit power of the UAV
$P_{d(i)}[t]$	Power allocation to the $d(i)$ -th GU in time slot t
$q^{\text{U}}[t]$	Location of the UAV in time slot t
$q_n^{\text{G}}, q_k^{\text{E}}, q^{\text{I}}$	Locations of the n -th GU, k -th eavesdropper, and IRS, respectively
r^{a}	Rician model coefficient
$R_{k \rightarrow d(i)}^{\text{UE}}[t]$	Eavesdropping rate of the k -th eavesdropper on the $d(i)$ -th GU in time slot t
$R_{d(i)}^{\text{sec}}[t], R^{\text{sec}}[t]$	Secrecy rate of the $d(i)$ -th GU and all GUs in time slot t , respectively
$t, T, \mathcal{T}$	Index, total number, and set of time slots
t_d	Duration of each time slot
α_0	Channel gain at the reference distance $d_0 = 1$ m
ρ_1, ρ_2	Path loss coefficient in $H^{\text{UR}}[t]$ and $H_n^{\text{RG}}[t]$, respectively
$\theta^{\text{UR}}[t], \xi^{\text{UR}}[t]$	Vertical and horizontal AoAs at the IRS in time slot t , respectively
$\gamma_{d(i)}[t]$	SINR of the $d(i)$ -th GU in time slot t
σ^2	Noise power

the elements in the system is described by using the 3D Cartesian coordinate system. Specifically, the location of the UAV in time slot t is represented by $q^{\text{U}}[t] = [x^{\text{U}}[t], y^{\text{U}}[t], H^{\text{U}}]$. Likewise, the locations of n -th GU, k -th eavesdropper, and IRS are denoted as $q_n^{\text{G}} = [x_n^{\text{G}}, y_n^{\text{G}}, 0]$, $q_k^{\text{E}} = [x_k^{\text{E}}, y_k^{\text{E}}, 0]$, $q^{\text{I}} = [x^{\text{I}}, y^{\text{I}}, z^{\text{I}}]$, respectively. Major parameters presented in the system model are summarized in Table II.

The subsequent presentation details the communication model and UAV mobility model of the considered system to characterize the main controllable variables and concerned objectives.

B. Channel Model

The communication channel between the UAV and each ground receiver consists of two parts that are the UAV transmit

link (*i.e.*, UAV-IRS link) and IRS reflection link (*i.e.*, IRS-GU link and IRS-Eavesdropper link), respectively.

1) *UAV-IRS Link*: The UAV-IRS link is modeled as a LoS channel due to the relatively high deployment altitude of both the UAV and IRS [28]. As such, let $H^{\text{UR}}[t] \in \mathbb{C}^{M_r M_c \times 1}$ represent the channel response vector between the UAV and IRS, which is denoted by

$$H^{\text{UR}}[t] = \sqrt{\frac{\alpha_0}{(d^{\text{UR}}[t])^{\rho_1}}} g_L^{\text{UR}}[t], \quad (1)$$

where α_0 denotes the reference channel power corresponding to a distance of $d_0 = 1$ m and ρ_1 is the parameter related to the path loss. Moreover, $d^{\text{UR}}[t]$ denotes the distance from the UAV and IRS. In addition, $g_L^{\text{UR}}[t] \in \mathbb{C}^{M_r M_c \times 1}$ is the LoS component, which is given by [29]

$$g_L^{\text{UR}}[t] = e^{-j2\pi \frac{d^{\text{UR}}[t]}{\lambda}} \times \left[1, \dots, e^{-j2\pi(M_r-1) \frac{d_r \sin \theta^{\text{UR}}[t] \cos \xi^{\text{UR}}[t]}{\lambda}} \right]^T \otimes \left[1, \dots, e^{-j2\pi(M_c-1) \frac{d_c \sin \theta^{\text{UR}}[t] \sin \xi^{\text{UR}}[t]}{\lambda}} \right]^T, \quad (2)$$

where the vertical and horizontal angles-of-arrival (AoAs) relative to the IRS are denoted by $\theta^{\text{UR}}[t]$ and $\xi^{\text{UR}}[t]$, respectively. Moreover, $(\cdot)^T$ represents the transpose operation.

2) *IRS-GU Link and IRS-Eavesdropper Link*: Although the IRS can provide a strong deterministic reflection path for ground nodes through passive beamforming, the complex ground environment inevitably introduces additional multipath components. Therefore, the Rician channel model, which consists of a deterministic component and multiple random scattering components, can characterize the IRS-GU and IRS-eavesdropper channels accurately [29]. Denote $H_n^{\text{RG}}[t] \in \mathbb{C}^{M_r M_c \times 1}$ as the channel from the IRS to the n -th GU during time slot t , which can be expressed as

$$H_n^{\text{RG}}[t] = \sqrt{\frac{\alpha_0}{(d_n^{\text{RG}})^{\rho_2}}} \left(\sqrt{\frac{r^{\text{a}}}{r^{\text{a}} + 1}} g_{n,L}^{\text{RG}}[t] + \sqrt{\frac{1}{r^{\text{a}} + 1}} g_{n,N}^{\text{RG}}[t] \right), \quad (3)$$

where r^{a} represents the Rician coefficient, and $g_{n,N}^{\text{RG}}[t] \sim \mathcal{CN}(0, \mathbf{I}_{M_r M_c})$ represents the small-scale fading modeled as a complex Gaussian random variable. Moreover, d_n^{RG} is the distance from the IRS and the n -th GU. Furthermore, $g_{n,L}^{\text{RG}}[t] \in \mathbb{C}^{M_r M_c \times 1}$ is the LoS component similar to Eq. (2). Likewise, the reflecting channel from the IRS to k -th eavesdropper, $H_k^{\text{RE}}[t] \in \mathbb{C}^{M_r M_c \times 1}$, is characterized using the same model.

C. NOMA-Based Communication Model

In time slot t , the IRS reflection phase coefficient matrix is given by

$$\Phi^{\text{I}}[t] = \text{diag}(\phi[t]) \in \mathbb{C}^{M_r M_c \times M_r M_c}, \quad (4)$$

where $\phi[t] = [e^{j\phi^{1,1}[t]}, \dots, e^{j\phi^{m_r, m_c}[t]}, \dots, e^{j\phi^{M_r, M_c}[t]}]^T \in \mathbb{C}^{M_r M_c \times 1}$. Given hardware restrictions, the IRS phase shifts are modeled as discrete quantities, *i.e.* $\phi^{m_r, m_c}[t] \in \left\{ 0, \frac{2\pi \cdot 1}{2^b}, \dots, \frac{2\pi \cdot (2^b - 1)}{2^b} \right\}$, where b indicates the number of quantization levels [30].

Consequently, in time slot t , the channels linking the UAV to the n -th GU and the k -th eavesdropper are given by

$$H_n^{\text{UG}}[t] = (H_n^{\text{RG}}[t])^T \Phi^{\text{I}}[t] H^{\text{UR}}[t], \quad (5)$$

$$H_k^{\text{UE}}[t] = (H_k^{\text{RE}}[t])^T \Phi^{\text{I}}[t] H^{\text{UR}}[t]. \quad (6)$$

In the considered system, the UAV employs NOMA to simultaneously serve N_{G} GUs, with the decoding order updated dynamically based on channel conditions. In downlink NOMA, power is allocated inversely to channel gain, *i.e.*, GUs with stronger channels receive less power [31]. As such, we rank the channel qualities of all GUs in descending order, which can be described as

$$H_{d(1)}^{\text{UG}}[t] > H_{d(2)}^{\text{UG}}[t] > \dots > H_{d(i)}^{\text{UG}}[t] > \dots > H_{d(N_{\text{G}})}^{\text{UG}}[t], \quad (7)$$

where $d(i)$ indicates the user index corresponding to the i -th position in the channel quality ranking. Therefore, the power allocation constraint can be expressed as

$$P_{d(1)}[t] < P_{d(2)}[t] < \dots < P_{d(i)}[t] < \dots < P_{d(N_{\text{G}})}[t], \quad (8)$$

where $\sum_{i=1}^{N_{\text{G}}} P_{d(i)}[t] = P^{\text{U}}[t]$, and $P^{\text{U}}[t]$ represents the transmit power of the UAV in time slot t .

In the sequential interference cancellation (SIC) process, $d(i)$ -th GU ($i < N_{\text{G}}$) can successively remove interference from GUs $d(i+1)$ to $d(N_{\text{G}})$ due to its stronger channel condition. The last user $d(N_{\text{G}})$ is unable to cancel interference from other signals [31]. Notably, the key decision parameters (*i.e.*, decoding order and power allocation) required for each GU to perform SIC are centrally calculated and distributed by the UAV, without requiring the GUs to estimate the channel state information (CSI) of other users for determining these decision parameters. Moreover, the SIC process neither causes privacy leakage nor conflicts with the anti-eavesdropping objective in our considered system due to the legitimacy of all GUs. Based on the SIC process, the signal-to-interference-plus-noise ratio (SINR) for the $d(i)$ -th GU in time slot t is expressed as

$$\gamma_{d(i)}[t] = \begin{cases} \frac{P_{d(i)}[t] (H_{d(i)}^{\text{UG}}[t])^2}{\sum_{j=1}^{i-1} P_{d(j)}[t] (H_{d(j)}^{\text{UG}}[t])^2 + \sigma^2}, & 1 < i \leq N_{\text{G}}, \\ \frac{P_{d(i)}[t] (H_{d(i)}^{\text{UG}}[t])^2}{\sigma^2}, & i = 1, \end{cases} \quad (9)$$

where σ^2 represents the noise power. Therefore, in time slot t , the communication rate for the $d(i)$ -th GU served by the UAV is expressed as $R_{d(i)}^{\text{UG}}[t] = \log_2(1 + \gamma_{d(i)}[t])$.

Moreover, the eavesdropping rate of the k -th eavesdropper on the $d(i)$ -th GU in time slot t is represented as [32]

$$R_{k \rightarrow d(i)}^{\text{UE}}[t] = \log_2 \left(1 + \frac{P_{d(i)}[t] (H_k^{\text{UE}}[t])^2}{\sum_{j=1, j \neq i}^{N_{\text{G}}} P_{d(j)}[t] (H_k^{\text{UE}}[t])^2 + \sigma^2} \right). \quad (10)$$

Therefore, in time slot t , the secrecy rate for the $d(i)$ -th GU is represented as [33]

$$R_{d(i)}^{\text{sec}}[t] = \left[R_{d(i)}^{\text{UG}}[t] - \max_{k \in \mathcal{K}} R_{k \rightarrow d(i)}^{\text{UE}}[t] \right]^+, \quad (11)$$

where $[\cdot]^+ \triangleq \max(0, \cdot)$. Accordingly, the secrecy rate of all GUS in time slot t can be denoted as

$$R^{\text{sec}}[t] = \sum_{i=1}^{N_{\text{G}}} R_{d(i)}^{\text{sec}}[t]. \quad (12)$$

D. UAV Mobility and Energy Models

In time slot t , the location of the UAV can be expressed as

$$x^{\text{U}}[t] = x^{\text{U}}[0] + \sum_{t'=1}^t a^{\text{x}}[t'], \quad y^{\text{U}}[t] = y^{\text{U}}[0] + \sum_{t'=1}^t a^{\text{y}}[t'], \quad (13)$$

where $x^{\text{U}}[0]$ and $y^{\text{U}}[0]$ represent the initial horizontal locations of the UAV. Moreover, $a^{\text{x}}[t] \in [-x_{\text{max}}, x_{\text{max}}]$ and $a^{\text{y}}[t] \in [-y_{\text{max}}, y_{\text{max}}]$ represent the distances covered by the UAV along the horizontal directions.

In our considered system, we introduce a rotary-wing UAV model. As such, the propulsion energy consumption of the UAV in time slot t is denoted as

$$e[t] = \left(P_{\text{s}} \left(1 + 3 \left(\frac{v^{\text{h}}[t]}{U_{\text{r}}} \right)^2 \right) + P_{\text{m}} \left(\sqrt{1 + \frac{1}{4} \left(\frac{v^{\text{h}}[t]}{V_{\text{h}}} \right)^4} - \frac{1}{2} \left(\frac{v^{\text{h}}[t]}{V_{\text{h}}} \right)^2 \right)^{\frac{1}{2}} + \frac{1}{2} d_0 \rho_{\text{a}} z G (v^{\text{h}}[t])^3 \right) t_{\text{d}}, \quad (14)$$

where $v^{\text{h}}[t] = \sqrt{(a^{\text{x}}[t])^2 + (a^{\text{y}}[t])^2} / t_{\text{d}}$ denotes the horizontal velocity in the time slot t . The meaning of other parameters can be found in [28].

E. Problem Formulation and Analysis

Let $\mathcal{Q} = \{q^{\text{U}}[t] \mid \forall t \in \mathcal{T}\}$, $\mathcal{P} = \{P_n \mid \forall n \in \mathcal{N}, t \in \mathcal{T}\}$, and $\Phi = \{\phi[t] \mid \forall t \in \mathcal{T}\}$ denote the UAV trajectory, power allocation for GUs, and IRS reflection phase coefficient matrix, respectively. According to the optimization variables defined above, the corresponding optimization objectives are presented as follows:

Objective 1: Considering that data leakage may lead to severe security risks, the main optimization objective is to maximize the average secrecy rate, which can be expressed as

$$f_1(\mathcal{Q}, \mathcal{P}, \Phi) = \frac{1}{T} \sum_{t=1}^T R^{\text{sec}}[t]. \quad (15)$$

Objective 2: Considering that excessive UAV energy consumption can shorten the operational duration of the system, another optimization objective focuses on reducing UAV energy consumption, which is given by

$$f_2(\mathcal{Q}) = \frac{1}{T} \sum_{t=1}^T e[t]. \quad (16)$$

As such, we formulate the IUSCMOP as follows:

$$\mathcal{P}1 : \max_{\mathcal{Q}, \mathcal{P}, \Phi} (f_1, -f_2), \quad (17a)$$

$$\text{s.t.} : X_{\min} \leq x^U[t] \leq X_{\max}, \quad \forall t, \quad (17b)$$

$$Y_{\min} \leq y^U[t] \leq Y_{\max}, \quad \forall t, \quad (17c)$$

$$\|a^x[t]\| \leq x_{\max}, \quad \forall t, \quad (17d)$$

$$\|a^y[t]\| \leq y_{\max}, \quad \forall t, \quad (17e)$$

$$\phi^{m_r, m_c}[t] \in \left\{ 0, \frac{2\pi \cdot 1}{2^b}, \dots, \frac{2\pi \cdot (2^b - 1)}{2^b} \right\}, \quad \forall t, \quad (17f)$$

$$P_{\min} \leq P^U[t] \leq P_{\max}, \quad \forall t \in \mathcal{T}, \quad (17g)$$

$$P_{d(1)}[t] < P_{d(2)}[t] < \dots < P_{d(N_G)}[t] \quad \forall t \in \mathcal{T}, \quad (17h)$$

where (17b) and (17c) guarantee that the UAV stays inside the specified target area. Moreover, (17d) and (17e) limit how far the UAV can travel in each time slot. In addition, (17f) specifies that the phase shift of the IRS should be determined from their predefined set. Furthermore, (17g) denotes the constraint of UAV transmit power, and (17h) represents the transmit power allocation to satisfy the NOMA constraint.

The key characteristics of the formulated IUSCMOP are outlined as follows:

(i) *IUSCMOP is a dynamic problem.* In the considered system, the UAV continuously updates its position, leading to variations in energy consumption and secrecy rate due to changing channel conditions and power allocation. Therefore, the formulated IUSCMOP is dynamic.

(ii) *IUSCMOP is a non-convex problem.* The IUSCMOP entails the joint optimization of multiple coupled variables, which introduces complex nonlinear relationships. Moreover, the discrete IRS phase shifts specified by constraint (17f) render the feasible region a non-convex set. Therefore, the IUSCMOP exhibits inherent non-convexity.

(iii) *IUSCMOP is an NP-hard problem.* To simplify the optimization problem, we initially concentrate solely on the primary objective while keeping the IRS phase shifts and UAV trajectory fixed, followed by equally allocating the UAV transmit power to all GUs. As such, the original IUSCMOP is simplified as

$$\mathcal{P}2 : \max_{\mathcal{P}} f_1, \quad (18a)$$

$$\text{s.t.} : P^U[t] \in \{0, P_{\max}\}, \quad \forall t, \quad (18b)$$

$$\sum_{t=1}^T P^U[t] \leq TP_{\max}, \quad (18c)$$

Clearly, the simplified optimization problem $\mathcal{P}2$ constitutes a knapsack problem and has been shown to be NP-hard [34]. Therefore, the original IUSCMOP is likewise NP-hard.

IV. PROPOSED APPROACH

In this section, we propose an agentic AI approach.

A. Motivations of Using Agentic AI Approach

Some conventional optimization algorithms, such as convex approximation and swarm intelligence algorithms, have been

applied to wireless optimization problems because of their comparatively low computational demands and straightforward implementation. However, they exhibit several limitations when applied to the IUSCMOP.

- *Over-reliance on Accurate Prior Knowledge:* These algorithms typically rely on complete prior information about the system state to achieve effective optimization. However, obtaining such prior knowledge in advance is challenging in wireless environments that are inherently dynamic and uncertain [31].
- *Lack of Adaptability:* These algorithms typically cannot adapt to environmental changes, which forces repeated iterations and retraining in dynamic systems.

Although DRL algorithms are suitable to solve dynamic optimization problems in wireless systems due to their adaptability, they still face the following challenges.

- *High Dimensionality of the Hybrid Action Space:* IRS is typically composed of numerous reflecting elements, which leads to a high-dimensional action space involving discrete IRS phase shifts. In this case, directly optimizing the IRS configuration using DRL may lead to slower convergence and degraded training accuracy [35].
- *Highly Dynamic and Complex Environment:* The multi-layer perceptron (MLP) is commonly used as the actor network in conventional DRL algorithms. However, the constrained modeling capability of the standard MLP limits the expressiveness of the actor network in highly dynamic and uncertain systems, which in turn hampers effective policy learning. Reference [36].

B. Overview of Proposed AO-GDMDDPG Approach

Different from the aforementioned conventional optimization algorithms, agentic AI that follows the perception-reasoning-action loop can autonomously explore and adapt to dynamic wireless environments without complete prior knowledge [37]. Moreover, agentic AI enables complex tasks to be addressed through specialized agents that handle individual sub-tasks efficiently [38]. These characteristics above make the agentic AI approach more suitable for solving the IUSCMOP than the aforementioned algorithms.

As such, we propose an agentic AI approach, namely the AO-GDMDDPG approach, which exhibits the following characteristics.

- *Multi-agent Structure:* To address the high-dimensional hybrid decision space in the IUSCMOP, AO-GDMDDPG adopts a multi-agent architecture, where each agent optimizes a specific type of decision variable. Specifically, the UAV agent optimizes the UAV trajectory and power allocation using GDMDDPG, while the IRS agent optimizes the IRS phase shifts using the AO method. Therefore, the proposed framework avoids handling the hybrid decision space with a single agent, thereby reducing the search complexity and improving the convergence speed and solution quality.
- *Multi-agent Collaboration based on Centralized Orchestration and Sequential Scheduling:* To coordinate heterogeneous agents, AO-GDMDDPG introduces a centralized

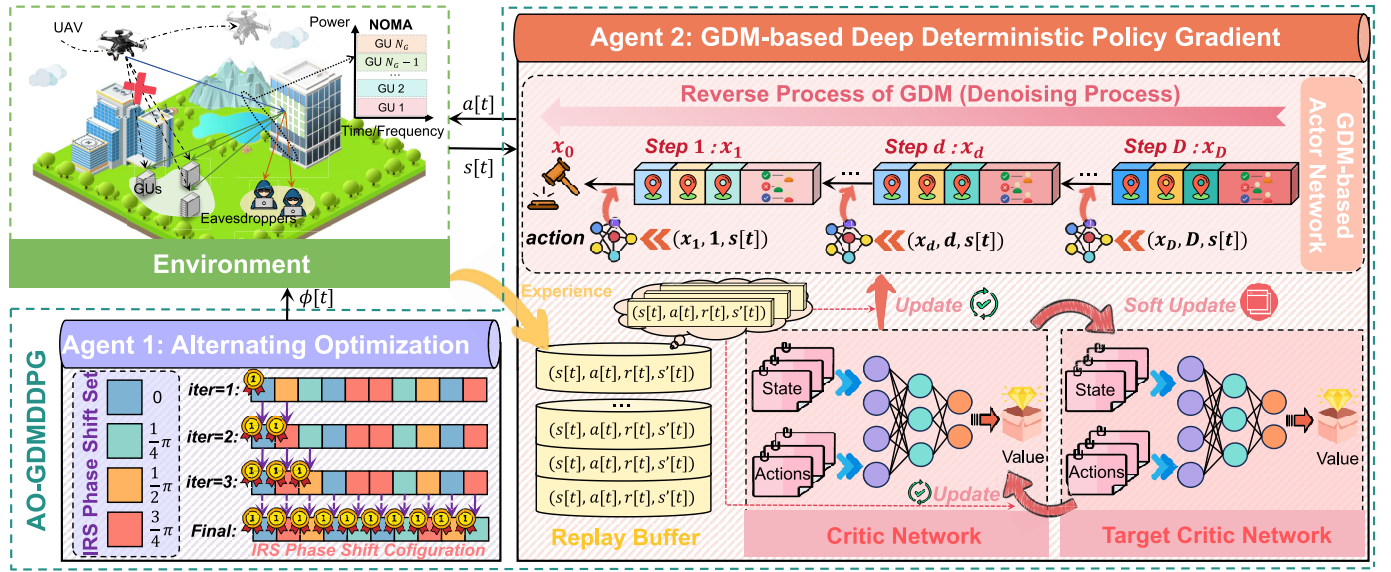


Fig. 2. The structure of the proposed AO-GMDDPG approach. Specifically, the observed environment state is input into the GMDDPG method to generate the UAV location and power allocation in the next time slot. Based on this, the IRS phase shifts are determined by using the AO method.

orchestration layer for task decomposition and scheduling. Specifically, the orchestration layer decomposes the IUSCMOP into multiple sub-tasks and assigns them to dedicated agents. Moreover, a sequential scheduling mechanism coordinates the optimization process among agents. Consequently, the proposed framework ensures effective collaboration without direct inter-agent communication, thereby improving the stability and convergence performance.

- *Powerful Environment State Analysis Capability and Improved Decision Quality:* GDM can effectively capture the latent distribution of data. Therefore, we integrate GDM into the UAV agent to overcome the limited representation capability of MLP-based actor networks. Specifically, the environment state is used as the conditional input of GDM to capture the underlying features of the dynamic environment, thereby guiding the reverse denoising process to generate high-quality decisions [39].

The workflow of the AO-GMDDPG approach is described as follows. First, the orchestration layer perceives the environment state from the dynamic wireless environment at the beginning of each time slot. Then, the orchestration layer sequentially schedules the UAV agent and IRS agent to perform corresponding inference processes. Specifically, the UAV agent treats the environment state as the conditional information for the reverse denoising process of GDM, thereby generating the UAV trajectory and power allocation through multi-step inference. Subsequently, based on the environment state and the decisions generated by the UAV agent, the IRS agent determines the IRS phase shift configurations through the AO method-based inference process. After the reasoning phase above is completed, the agents execute their decisions, and this process directly influences the physical wireless environment. In turn, the environment returns feedback rewards according to the system performance and completes the transition of the environment state.

Notably, the proposed AO-GMDDPG approach differs from the multi-agent DRL algorithms and traditional hierarchical optimization algorithms. Specifically, compared with standard multi-agent DRL, where homogeneous agents independently perform direct state-action mapping [40], the proposed AO-GMDDPG approach employs heterogeneous dedicated agents, GDM-based generative reasoning, and centralized orchestration-based collaboration, thereby achieving more effective coordination and higher-quality decision-making in dynamic wireless environments. Moreover, traditional hierarchical optimization methods rely on predefined mathematical models and fixed solution procedures [41]. In contrast, the AO-GMDDPG approach can autonomously learn and evolve in dynamic wireless communication scenarios through continuous experience accumulation and policy updates accordingly, ultimately achieving high-quality real-time decision-making capabilities.

Fig. 2 and Algorithm 1 illustrate the architecture and pseudocode of the AO-GMDDPG approach, and the subsequent sections describe the comprehensive design of its two agents.

C. UAV Agent: GMDDPG-Based Continuous Variable Optimization

This part begins by introducing the Markov decision process (MDP) and subsequently offers a comprehensive explanation of the proposed GMDDPG method.

1) *MDP:* The MDP serves as the mathematical foundation of DRL algorithms, which consists of a state space $\mathcal{S}$, an action space $\mathcal{A}$, a reward function $\mathcal{R}$, a discount factor γ , and a state transition function. Notably, the first three components are essential and can be defined based on IUSCMOP as follows:

- **State $\mathcal{S}$:** Considering that the UAV can observe its own location, the state $s[t] \in \mathcal{S}$ in time slot t can be given by

$$s[t] = [t, q^U[t], q^G[t], q^E[t], q^I], \quad (19)$$

where t represents the index of the current time slot. Moreover, $q^G[t] = \{q_n^G[t] | n \in \mathcal{N}\}$ and $q^E[t] = \{q_k^E[t] | k \in \mathcal{K}\}$

Algorithm 1 AO-GMDDPG Approach

```

1 Initialize the parameters of the actor network and critic
  network, i.e.,  $\pi_\varphi$  and  $Q_\eta$ , respectively. Initialize the
  target networks  $\pi_{\varphi'}$  and  $Q_{\eta'}$  with  $\varphi' \leftarrow \varphi$  and
   $\eta' \leftarrow \eta$ .
2 Initialize the experience replay buffer  $\mathcal{B}$  and maximum
  number of episodes  $E_p$ .
3 for episode = 1 to  $E_p$  do
4   for  $t = 1$  to  $T$  do
5     Observe state  $s[t]$  and sample  $x_D \sim \mathcal{N}(0, \mathbf{I})$ .
6     for  $d = D$  to 0 do
7       Construct denoising network  $\epsilon(x_d, d, s[t])$ .
8       Get  $\mu_\varphi$  and  $\bar{\beta}_d$  based on Eqs. (25) and (24).
9       Get  $x_d$  according to Eq. (26).
10      Obtain the action  $a[t]$  based on the above
        denoising iterations.
11      Calculate the IRS phase shifts  $\Phi[t]$  according
        to Algorithm 2.
12      Execute  $a[t]$  and  $\Phi[t]$ , obtain the reward  $r[t]$ 
        and the next state  $s'[t]$ .
13      Restore the experience tuple in  $\mathcal{B}$ .
14      Sample  $B$  experiences in  $\mathcal{B}$ , and use them to
        update the networks according to Eqs. (27)
        and (28).
15      Update target networks according to Eqs. (29).
16 Return the actor network  $\pi_\varphi$ .
```

denote the location set of all legitimate GUs in time slot t and eavesdroppers, respectively.

- **Action $\mathcal{A}$:** Based on the observed state $s[t]$, the agent selects an action $a[t] \in \mathcal{A}$ that encompasses the continuous optimization variables. In this case, $a[t]$ is defined as

$$a[t] = [a^x[t], a^y[t], P[t]], \quad (20)$$

where $P[t] = \{P_n[t] | n \in \mathcal{N}\}$ denotes the set of transmit power allocated to all GUs.

- **Reward $\mathcal{R}$:** A reward function that closely aligns with the optimization objectives offers the agent a well-defined direction for improvement. In the field of multi-objective optimization, designing a linear weighted sum-based reward function is a common and effective approach. As such, the reward $r_t \in \mathcal{R}$ is designed as

$$r[t] = \omega_{\text{sec}} R^{\text{sec}}[t] + \omega_e e[t] + r^{\text{out}}[t], \quad (21)$$

where ω_{sec} and ω_e are constants used to scale the two optimization objectives to the same order of magnitude, thereby effectively eliminating the influence of physical units so that the reward value becomes dimensionless. In addition, $r^{\text{out}}[t]$ is expressed as

$$r^{\text{out}}[t] = \begin{cases} 0, & \text{if } x^U[t] \in [X_{\min}, X_{\max}] \text{ and} \\ & y^U[t] \in [Y_{\min}, Y_{\max}], \\ \mathbf{p}_1, & \text{otherwise} \end{cases} \quad (22)$$

where $\mathbf{p}_1 < 0$ represents the penalty applied if the UAV moves beyond the designated target region.

2) *GMDDPG Method:* The GDM operates through two complementary stages. Initially, the forward stage incrementally introduces noise into the original data, ultimately transforming it into complete noise. Subsequently, the reverse stage reconstructs the data distribution in a stepwise manner, recovering the original data from the noisy representation. It is worth noting that the original data is typically treated as the optimal solution in the optimization problems [36]. However, obtaining the optimal solution of the IUSCMOP beforehand is impractical. Moreover, the modeling and reasoning capability of GDM primarily comes from its reverse stage. As such, we only employ the reverse stage to assist the DDPG algorithm in determining actions based on the environment states.

To approximate the reverse stage of the GDM, a parameterized conditional probability distribution $p_\varphi(x_{d-1} | x_d)$ is employed to model the transition from the noise data x_d at step d to that of the previous step x_{d-1} , which is given by

$$p_\varphi(x_{d-1} | x_d) = \mathcal{N}(x_{d-1}; \mu_d = \mu_\varphi(x_d, d, s), \Sigma_d = \bar{\beta}_d \mathbf{I}), \quad (23)$$

where μ_φ is a mean function implemented by a DNN with parameter φ . Moreover, s represents the observed environment state. In addition, $\bar{\beta}_d$ represents the variance, which is denoted as

$$\bar{\beta}_d = \frac{1 - \bar{\alpha}_{d-1}}{1 - \bar{\alpha}_d} \beta_d, \quad (24)$$

where β_d is calculated by a noise schedule function. Moreover, $\alpha_d = 1 - \beta_d$, and $\bar{\alpha}_d = \prod_{j=1}^d \alpha_j$.

According to the work [36], the mean μ_φ of the distribution can be expressed as

$$\mu_\varphi(x_d, d, s) = \frac{1}{\sqrt{\bar{\alpha}_d}} (x_d - \beta_d \sqrt{1 - \bar{\alpha}_d} \tanh(\varepsilon_\varphi(x_d, d, s))), \quad (25)$$

where ε_φ denotes the denoising network parameterized by φ and generates the denoising noise using the present environment state s .

As such, according to Eq. (23), the data x_{d-1} is drawn from the distribution $p(x_d) p_\varphi(x_{d-1} | x_d)$. Similarly, the final data sample x_0 is obtained by recursively applying this sampling process from step D to step 0, where D represents the number of denoising steps. However, when training generative models with inherent randomness, back-propagating gradients through random variables poses a common challenge [36]. To overcome these limitations, the reparameterization technique is employed to decouple the randomness from the distribution parameters, which is represented as

$$x_{d-1}[t] = \mu_\varphi(x_d, d, s[t]) + \frac{\bar{\beta}_d}{2} \odot \varepsilon, \quad (26)$$

where $\varepsilon \sim \mathcal{N}(0, \mathbf{I})$. Through repeated application of Eq. (26), the actor network π_φ produces the action $a[t]$ (i.e., $x_0[t]$) corresponding to the environment state $s[t]$.

Moreover, the critic network $Q_\eta(s, a)$ is employed to evaluate the quality of actions generated by the GDM-based actor network under the current environment state s , thereby providing essential guidance for policy optimization. To improve training stability, target networks $\pi_{\varphi'}(s)$ and $Q_{\eta'}(s, a)$ are introduced. In addition, the experience tuples

$(s[t], a[t], r[t], s'[t])$ produced throughout training are saved in an experience buffer $\mathcal{B}$ to facilitate network updates.

Therefore, using B experiences extracted from the experience buffer $\mathcal{B}$ updates the critic network, and the loss function can be expressed as

$$\mathcal{L}(\eta) = \frac{1}{B} \sum_{b=1}^B (y[b] - Q_{\eta}(s[b], a[b]))^2, \quad (27)$$

where $y[b]$ represents the target value.

Similarly, the GDM-based actor network is updated based on the loss function defined as

$$\mathcal{L}(\varphi) = -\frac{1}{B} \sum_{b=1}^B Q_{\eta}(s[b], \pi_{\varphi}(s[b])). \quad (28)$$

Moreover, updates to the target networks are performed as follows:

$$\eta' \leftarrow \xi\eta + (1 - \xi)\eta', \quad \varphi' \leftarrow \xi\varphi + (1 - \xi)\varphi', \quad (29)$$

where ξ is the weight factor of the soft-update process.

D. IRS Agent: AO-Based Discrete Variable Optimization

The discrete IRS phase configuration affects the secrecy rate of the system and is independent of the UAV energy consumption. As such, when the UAV location and power allocation are fixed, the IUSCMOP can be reformulated as a secrecy rate maximization problem that optimizes only the IRS phase shifts, which is expressed as

$$\mathcal{P3} : \max_{\Phi} \frac{1}{T} \sum_{t=1}^T R^{\text{sec}}[t], \quad (30a)$$

$$\text{s.t.} : \phi^{m_r, m_c}[t] \in \left\{0, \frac{2\pi \cdot 1}{2^b}, \dots, \frac{2\pi \cdot (2^b - 1)}{2^b}\right\}, \forall t. \quad (30b)$$

Algorithm 2 AO Method

Input: UAV location $q^U[t]$ and power allocation $P[t]$.

Output: Phase shifts of IRS $\phi[t]$.

- 1 Randomly initialize the phase shifts from $\phi \in \{0, \frac{2\pi \cdot 1}{2^b}, \dots, \frac{2\pi \cdot (2^b - 1)}{2^b}\}$.
- 2 Initialize the maximum number of iterations $\text{Iter}_{\max}$.
- 3 **for** $iter = 1$ to $\text{Iter}_{\max}$ **do**
- 4 **for** $r = 1$ to M_r **do**
- 5 **for** $c = 1$ to M_c **do**
- 6 **for** $\phi^{r,c}[t]$ in $\{0, \frac{2\pi \cdot 1}{2^b}, \dots, \frac{2\pi \cdot (2^b - 1)}{2^b}\}$ **do**
- 7 $\phi^{r',c}[t]$ is fixed, $\forall r' \neq r, c' \neq c$.
- 8 Calculate the channel gain and determine the power allocation.
- 9 Calculate the secrecy rate of all GUs.
- 10 Set $\phi^{r,c}[t]$ when $R^{\text{sec}}[t]$ is maximized.

As presented in Algorithm 2, a straightforward yet effective AO method is utilized for addressing the simplified optimization problem [42]. *First*, the AO method initializes the phase shifts of all IRS reflecting elements by using a predefined discrete phase shift set. *Then*, a per-element optimization strategy

is adopted, where each reflecting element is optimized individually while keeping the others fixed. Specifically, for a given reflecting element, the AO method traverses each candidate phase shift in the discrete set to calculate the corresponding channel gain between the UAV and GUs. According to the calculated channel gain, the power allocation is determined using Eq. (17h), and the total secrecy rate of all users under each candidate is then obtained. The phase shift yielding the highest total secrecy rate is chosen as the final setting for the specific reflecting element. *Finally*, this procedure is then iterated until the predefined maximum number of iterations is completed.

E. Computation Complexity and Deployment Feasibility Analysis

1) *Computation Complexity:* The computation complexity of the AO-GDMDDPG approach consists of the following two parts.

Computation Complexity of AO Method: Let the maximum number of iterations for the AO method be denoted as $\text{Iter}_{\max}$, the computational complexity of the AO method can be expressed as $\mathcal{O}(\text{Iter}_{\max} \times M_r \times M_c \times 2^b)$.

Computation Complexity of GDMDDPG Method: First, let $|\varphi|$ and $|\eta|$ denote the number of parameters in the actor and critic networks, respectively. In this case, the complexity of the initialization phase is $\mathcal{O}(2|\varphi| + 2|\eta|)$. Next, the total number of training episodes is E_p , and the computational complexity of the action selection stage is $\mathcal{O}(E_p \times T \times D \times |\varphi|)$. Then, let V represent the complexity of agent-environment interaction, and the experience collection phase has a complexity of $\mathcal{O}(E_p \times T \times V)$. Finally, the network update phase incurs a complexity of $\mathcal{O}(E_p \times T \times (2|\varphi| + 2|\eta|))$ [39].

In summary, the computation complexity of the AO-GDMDDPG approach can be expressed as $\mathcal{O}(2|\varphi| + 2|\eta| + E_p \times T \times (\text{Iter}_{\max} \times M_r \times M_c \times 2^b + D|\varphi| + V + 2|\varphi| + 2|\eta|))$.

Remark 2: The computation complexity of the AO method for large-scale IRS remains controllable. First, the optimization complexity can be reduced by grouping adjacent reflecting elements. Specifically, due to spatial correlation in wireless channels, adjacent elements tend to exhibit similar optimal responses. Inspired by [43], L adjacent elements can share the same phase shift, thereby reducing the optimization dimension from $M_r \times M_c$ to $M_r \times M_c / L$. Moreover, practical IRS systems generally employ low-resolution phase shifts due to hardware constraints, which further limits complexity growth. Existing works commonly adopt no more than 2-bit phase shift quantization [44], while [42] has demonstrated that 2-bit quantization can already achieve satisfactory communication performance. Therefore, the computation complexity of the AO method remains controllable in practical deployments.

2) *Deployment Feasibility Analysis:* AO-GDMDDPG supports real-time implementation due to the separation between computationally intensive offline training and lightweight online deployment. Specifically, the large-scale parameter updates and network optimization process are completed offline at ground base stations or cloud servers with strong computing capability to obtain a well-trained GDMDDPG. After the training stage, the UAV agent only performs

TABLE III
SIMULATION PARAMETERS

Notation	Value	Notation	Value
$X_{\max}$	500 m	$Y_{\max}$	500 m
$x_{\max}$	20 m	$y_{\max}$	20 m
t	1 s	d_r	$\frac{\lambda}{2}$
d_c	$\frac{\lambda}{2}$	α_0	10^{-3}
ρ_1	2	ρ_2	2.2
r^a	1	M_r/M_c	4~12
σ^2	-110 dbm	T	80 ~ 120

lightweight forward propagation on the trained GDMDDPG, while the IRS agent executes a simple deterministic AO search, which results in relatively low online inference latency. Therefore, such a hybrid training-deployment architecture enables agents to respond quickly and supports real-time implementation.

V. SIMULATIONS

This section presents simulation results that illustrate the superior performance of our proposed approach.

A. Simulation Setups

In this part, we provide the simulation setups and show the comparison algorithms.

1) *Environment Details*: A target area measuring 500 m $\times$ 500 m is considered, within which a UAV operates and communicates with five randomly located GUs while two eavesdroppers are present. Moreover, the UAV starts at [500 m, 0 m, 60 m] and flies at a fixed altitude of 60 m. In addition, an IRS is deployed at [0 m, 300 m, 50 m]. Other major parameters are summarized in Table III, and the parameters associated with the UAV energy consumption model are detailed in [45].

2) *Algorithm Parameters*: The critic network in GDMDDPG adopts a three-layer MLP, where each hidden layer contains 256 neurons with ReLU activation. The GDM-based actor network consists of a state encoder, a step encoder, and an output layer. The state encoder uses two linear layers with ReLU. The step encoder applies sinusoidal positional encoding to the denoising step index, followed by two linear layers with ReLU. The output layer concatenates the sampled action in the last step, the step index, and the encoded state, and processes them through three linear layers with activation to generate actions. Notably, the final linear layer in the output layer adopts the tanh activation function to constrain the generated actions within the range of $[-1, 1]$. This design facilitates subsequent linear scaling according to the predefined UAV displacement boundaries, thereby ensuring that the UAV displacement satisfies the constraints in Eqs. (17d) and (17e) during each time slot. The experience buffer is configured with a size of 10^6 and a mini-batch of 128. Optimization of both networks is performed using Adam, employing a discount factor of 0.99 and a soft update weight of 0.05.

3) *Comparison Algorithms*: The performance of the proposed approach is evaluated by employing two comparative strategies.

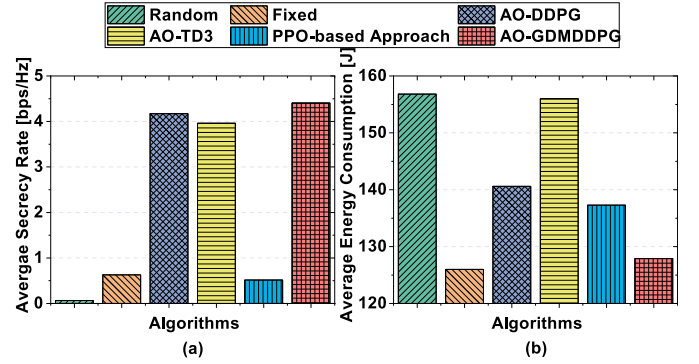


Fig. 3. Optimization objectives obtained by different algorithms. (a) Average secrecy rate. (b) Average UAV energy consumption.

- **Fixed Strategy**: In this strategy, the UAV travels at a fixed maximum endurance speed from its starting point to a hovering position above the IRS, with equal power distributed among GUs and all IRS reflecting elements set to a phase shift of $\frac{\pi}{2}$.
- **Random Strategy**: In this strategy, the IRS phase shifts, UAV trajectory, and power allocation for GUs are randomly set.

Moreover, the AO-GDMDDPG approach is compared against two DRL-based methods.

- **AO-DDPG**: The conventional DDPG algorithm is applied to design both the UAV trajectory and the power allocation among GUs [46]. Moreover, the optimization of the IRS phase shift is decided by the AO method.
- **AO-Twin Delayed Deep Deterministic Policy Gradient (TD3)**: TD3 is widely regarded as an improved version of DDPG that mitigates Q-value overestimation [47]. The TD3 approach is employed to determine both UAV trajectory planning and power allocation for GUs, while IRS phase shifts are simultaneously refined using the AO method.
- **PPO-based Approach**: This approach was proposed by [25]. We define the action space of the PPO-based approach to include all optimization variables so that it remains consistent with the core design of [25]. Moreover, we design the state space and reward function of the PPO-based approach to be consistent with our designed MDP to match the characteristics and requirements of our considered system.

B. Simulation Results

1) *Optimization Performance Evaluation*: Fig. 3 shows the optimization objectives achieved by different algorithms. It is evident that AO-GDMDDPG achieves higher average secrecy rate while reducing UAV energy consumption compared with AO-DDPG and AO-TD3. This performance improvement results from the enhancement of the action generation strategy through the reverse process of GDM. Specifically, in GDMDDPG, actions are initially sampled at random and then gradually refined through the actor network via denoising. Meanwhile, the critic network evaluates the generated actions and provides optimizing direction for the GDM-based actor network. This generation-evaluation collaborative

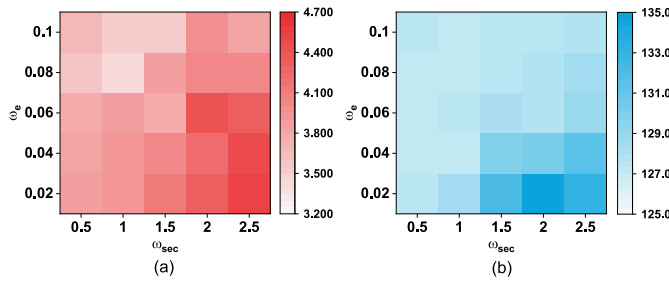


Fig. 4. Sensitivity analysis of the weight parameter in the reward function. (a) Average secrecy rate. (b) Average UAV energy consumption.

mechanism improves the ability of GMDDPG to express environment states and the effectiveness of actions. Note that the results achieved by the AO-GMDDPG approach show a gap from the theoretical optimal values. This is because the theoretical optimal value is essentially a performance upper bound under ideal conditions, which makes it difficult to achieve through approach training and optimization due to the dynamic characteristics of the environment and the trade-offs inherent in multi-objective optimization. Nevertheless, the AO-GMDDPG approach still approaches the theoretical optimal values more than other comparison algorithms. Moreover, the fixed strategy yields the lowest energy consumption, as the UAV maintains the maximum endurance speed. Furthermore, the random strategy performs worst in secure communication due to fully random optimization variables.

2) *Impact of Hyper-Parameters of AO-GMDPPG*: Fig. 4 presents the average secrecy rate and UAV energy consumption achieved by the AO-GMDDPG approach under different combinations of ω_{sec} and ω_e . Specifically, we vary ω_{sec} from 0.5 to 2.5 with a step size of 0.5 and vary ω_e from 0.02 to 0.1 with a step size of 0.02, thereby representing different importance levels of the secrecy rate and UAV energy consumption in the reward function, respectively. As we can see, when ω_{sec} is higher and ω_e is smaller, the AO-GMDDPG approach gives higher priority to the secrecy performance at the expense of UAV energy consumption. In contrast, when ω_e is higher, the AO-GMDDPG approach tends to make the UAV fly near the maximum-endurance speed to reduce the UAV energy consumption. The results above demonstrate a clear trade-off between the two optimization objectives and reveal that different weighting parameter settings can be set according to different system requirements. Since we aim to balance the secure communication performance and energy efficiency in our considered system, we set $\omega_{\text{sec}} = 2$ and $\omega_e = 0.06$ in all simulations.

Fig. 5(a) shows the convergence curves of the proposed AO-GMDDPG approach with different learning rates. The effectiveness of DRL depends heavily on the learning rate, as it governs both the speed and reliability of policy adjustments informed by reward signals. We can observe that setting the learning rate to 0.0005 causes pronounced oscillations during training, as substantial parameter updates disrupt the stability of policy learning. In contrast, a learning rate of 0.0001 leads to much slower convergence and lower final performance, as small updates make the algorithm prone to getting stuck in

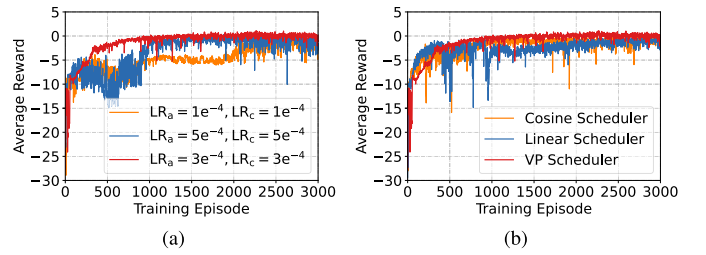


Fig. 5. Convergence of the proposed AO-GMDDPG approach with different parameters. (a) Learning rates. (b) Noise schedule functions.

TABLE IV
STATISTICAL ANALYSIS OF CONVERGENCE OF AVERAGE UAV ENERGY CONSUMPTION IN FIRST 500 EPISODES

Algorithms	Variance	Mean	Max	Min
PPO-based Approach	237.59	174.47	212.42	146.47
AO-TD3	606.95	177.35	248.69	151.23
AO-DDPG	421.53	167.38	227.10	139.80
AO-GMDDPG	121.62	140.92	171.22	126.89

TABLE V
INFERENCE TIME AND REWARD VALUE OF AO-DDPG AND AO-GMDDPG WITH DIFFERENT DENOISING STEPS D

Metric	AO-DDPG	AO-GMDDPG						
		$D=1$	$D=2$	$D=3$	$D=4$	$D=5$	$D=6$	$D=7$
Inference Time (s)	0.118	0.120	0.125	0.125	0.126	0.127	0.129	0.130
Reward Value	-1.744	-1.24	-1.09	-0.69	-0.81	-1.06	-1.07	-0.97

local optima. Therefore, the learning rate is set to 0.0003 to strike a balance between convergence speed and final accuracy.

Fig. 5(b) shows the convergence curves of the AO-GMDDPG approach under different noise schedule functions. The performance of AO-GMDDPG depends on the reverse process of GDM, which is governed by the noise schedule function. This function controls the temporal evolution of noise levels during the reverse process, thereby influencing the quality of generated actions. In this case, we compare three widely used noise functions, including the VP function, linear function, and cosine function. The corresponding results demonstrate that the VP-based approach converges more rapidly and delivers improved accuracy, which confirms its suitability for addressing the IUSCMOP.

Table V presents the average inference time for each decision process and reward values in the AO-DDPG approach and AO-GMDDPG approach with different numbers of denoising steps D . As can be seen, both too few and too many denoising steps lead to degraded performance. With too few steps, the generated actions retain substantial noise, which degrades their quality. On the other hand, too many steps cause over-smoothing, which weakens the exploration ability of the algorithm. As such, the number of denoising steps is set to 3 according to the reward values. Moreover, the inference time of AO-GMDDPG gradually increases with the number of denoising steps due to its increasing computation complexity. Nevertheless, we can find that when the optimal number of

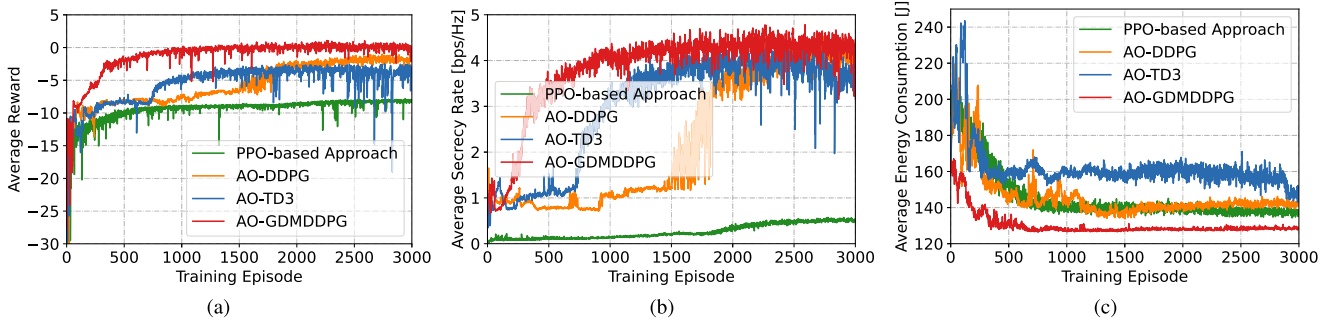


Fig. 6. Convergence of different algorithms. (a) Average reward. (b) Average secrecy rate. (c) Average UAV energy consumption.

denoising steps is 3, the inference time of AO-GDMDDPG only increases by about 6.2% compared to AO-DDPG, while the reward value increases by about 61.1%. This indicates that the additional computational complexity introduced by GDM is relatively small, whereas it achieves a significant performance improvement.

3) Algorithm Convergence Performance Evaluation:

Fig. 6(a) shows the reward curves of AO-DDPG, AO-TD3, and AO-GDMDDPG. All approaches exhibit notable training oscillations and low rewards during the initial phase. This phenomenon occurs because the agent frequently violates the constraints related to the UAV movement boundaries defined in the IUSCMOP during the exploration phase, which leads to large penalties. Moreover, although the PPO-based approach achieves satisfactory performance in [25], it converges to the lowest value in terms of both the average reward and average secrecy rate in our simulation environment. This phenomenon occurs because the existing work [25] assumes that the IRS contains only 16 reflecting elements, whereas our setting includes 64 elements. As such, the dimensionality of the action space becomes extremely high, which makes it difficult for the conventional PPO-based agent to explore effectively. In contrast, the proposed AO-GDMDDPG approach achieves faster convergence and better stability than the PPO-based approach. This is because the proposed approach employs two dedicated agents to reduce the optimization difficulty of the high-dimensional action space, while GDM improves the modeling accuracy of the state-action relationship. Therefore, the proposed approach can achieve more effective policy learning in dynamic environments. In addition, as shown in Table V and Fig. 6(a), the results demonstrate that the proposed AO-GDMDDPG approach still achieves superior performance compared to the baseline approaches even with a single-step or few-step reverse process. This indicates that reducing the number of denoising steps in GDM can control the additional complexity while not significantly affecting policy performance.

In addition, as shown in Figs. 6(b) and 6(c), AO-GDMDDPG outperforms AO-DDPG and AO-TD3 in both the average secrecy rate and UAV energy consumption. These results demonstrate that AO-GDMDDPG can enhance communication security while reducing UAV energy consumption, which is essential for an energy-limited UAV platform. Furthermore, we analyze the energy consumption stability of

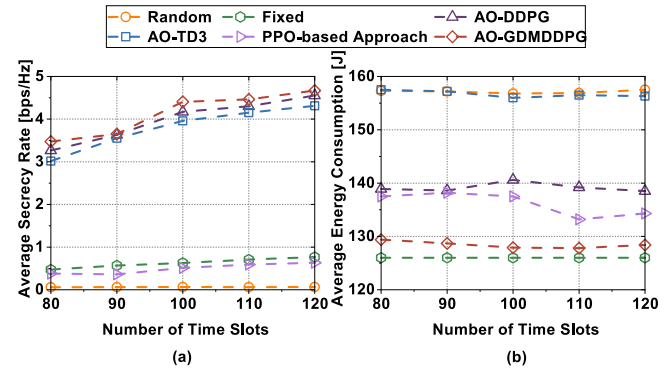


Fig. 7. The impact of time slot length on the optimization objectives. (a) Average secrecy rate. (b) Average UAV energy consumption.

different algorithms during the early training stage. Table IV presents the statistical results in the first 500 episodes of Fig. 6(c). The variance of AO-GDMDDPG is only 121.62, which is significantly lower than those of the PPO-based approach, AO-TD3, and AO-DDPG. Therefore, this result reflects that AO-GDMDDPG achieves more stable performance while maintaining effective exploration during early training.

C. Impact of System Settings

1) *Impact of Number of Time Slots*: Fig. 7 shows the average secrecy rate and UAV energy consumption with different numbers of time slots. Notably, the secrecy rate improves as the number of time slots increases. This phenomenon occurs because the UAV stays closer to the IRS for extended periods, enhancing the channel gain and thus improving the secrecy rate. Moreover, the secrecy rate of the fixed strategy is lower, mainly due to the limitations of the fixed IRS phase shift design. In addition, the random strategy yields a near-zero secrecy rate, since the UAV rarely moves close to the IRS, resulting in poor communication quality. Note that AO-GDMDDPG achieves a higher secrecy rate and lower energy consumption across time slot settings, demonstrating the effectiveness of AO-GDMDDPG in adapting to environmental changes and solving the formulated IUSCMOP. This performance improvement results from the accurate passive beamforming direction optimization through the AO method and improved action quality enabled by the GDM.

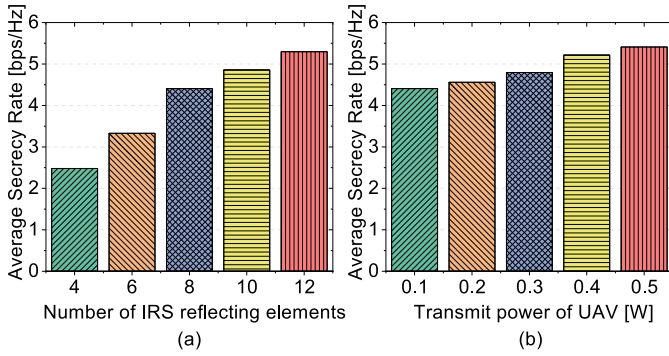


Fig. 8. Impact of system settings on the secrecy rate of the AO-GDMDDPG approach. (a) Impact of the number of IRS reflecting elements. (b) Impact of the transmit power of UAV.

2) Impact of Number of IRS Reflecting Elements:

Fig. 8(a) shows the average secrecy rate of the proposed AO-GDMDDPG approach under different numbers of IRS reflecting elements. We can find that the average secrecy rate of the system increases with a larger number of IRS reflecting elements. This is because more reflecting elements allow the IRS to more accurately direct the reflected signal, optimizing the transmission path and improving the channel quality for legitimate GUs, while also preventing eavesdropping. Moreover, Fig. 8(b) presents the secrecy performance variances under different transmit powers. As we can see, the secrecy performance improves as the transmit power increases, since a higher transmit power strengthens the legitimate communication links.

D. Impact of Imperfect CSI

In our considered system, eavesdroppers may remain covert in practical scenarios, which makes it challenging to obtain their accurate CSI in advance. In this special case, the channel estimation between the IRS and eavesdroppers is generally subject to measurement errors, and the corresponding channels are expressed as $\hat{H}_k^{\text{RE}}[t] = H_k^{\text{RE}}[t] + \Delta h_k[t]$, where $H_k^{\text{RE}}[t]$ represents the estimated channels for the IRS-eavesdropper k link, and $\Delta h_k[t]$ represents the corresponding CSI error [48]. Moreover, we denote $\Omega_k \triangleq \{\Delta h_k[t] \mid \|\Delta h_k[t]\| \leq \epsilon\}$ as the CSI error set, where $\epsilon = \delta \|H_k^{\text{RE}}[t]\|$ is the error bound and δ is the relative amount of CSI uncertainties [48]. Based on this channel model with imperfect CSI, the secrecy rate $R_{d(i)}^{\text{sec}}[t]$ for the $d(i)$ -th GU is represented as

$$R_{d(i)}^{\text{sec}}[t] = \left[R_{d(i)}^{\text{UG}}[t] - \max_{\substack{k \in \mathcal{K} \\ \Delta h_k[t] \in \Omega_k}} R_{k \rightarrow d(i)}^{\text{UE}}[t] \right]^+, \quad (31)$$

where $R_{d(i)}^{\text{UG}}[t]$ and $R_{k \rightarrow d(i)}^{\text{UE}}[t]$ represent the transmission rate of the $d(i)$ -th GU and k -th eavesdropper.

We further evaluate the performance of the proposed AO-GDMDDPG approach under channel estimation errors. Specifically, Fig. 9 presents the average secrecy rate of different algorithms under different levels of CSI error. As can be seen, the secrecy performance under imperfect CSI is lower than that under perfect CSI. This is because the IRS agent

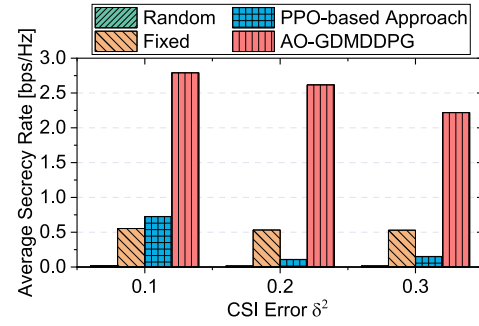


Fig. 9. Average secrecy rate versus different CSI error levels.

relies on CSI for phase shift configuration, which indicates that inaccurate CSI may cause the configuration to deviate from the theoretically optimal solution, thereby resulting in secrecy performance degradation. Notably, the simulation results show that the performance loss of the AO-GDMDDPG approach remains limited compared to other algorithms. This phenomenon indicates that although the proposed approach is affected by the CSI accuracy, it can still maintain stable and relatively favorable secrecy performance within a certain error range. Considering that the AO method currently adopted by the IRS agent is not very adaptive, more robust IRS beamforming methods can be integrated into the IRS agent to deal with imperfect CSI. For example, the authors in [49] proposed a Transformer-based passive beamforming method, which employs a self-attention mechanism to capture the structural characteristics of the wireless channel to learn IRS configurations. Since our proposed approach adopts a modular agentic AI architecture, such robust IRS beamforming methods can be directly incorporated into the IRS agent, thereby improving the performance under CSI error while preserving the overall approach architecture.

VI. CONCLUSION

In this paper, we have considered an IRS-assisted low-altitude UAV secure communication system, where a UAV employs NOMA to serve multiple legitimate GUs with the assistance of an IRS. Moreover, we have formulated the IUSCMOP to maximize the average secrecy rate and minimize the average UAV energy consumption by jointly optimizing the UAV trajectory, power allocation, and IRS phase shifts. To solve the IUSCMOP, we have proposed the AO-GDMDDPG approach, which integrates the IRS agent and UAV agent to separately optimize the discrete and continuous variables, which improves the learning efficiency and solution quality. Simulation results have verified that the AO-GDMDDPG approach delivers improved secure communication capability while preserving low UAV energy consumption. In particular, the proposed approach has achieved more stable performance than other comparison algorithms under different system settings, which indicates the robustness and adaptability of the proposed approach [49].

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